

The diagram shows a 12V LED driver circuit. It includes a 13V8 input, a MOSFET (Q1, NDS352AP), a diode (D2, 14V), a resistor (R1, 100 ohms, 100mW), and an LED (D1, MMSZ5244BT1G). The circuit is powered by a 12V0 source and grounded.

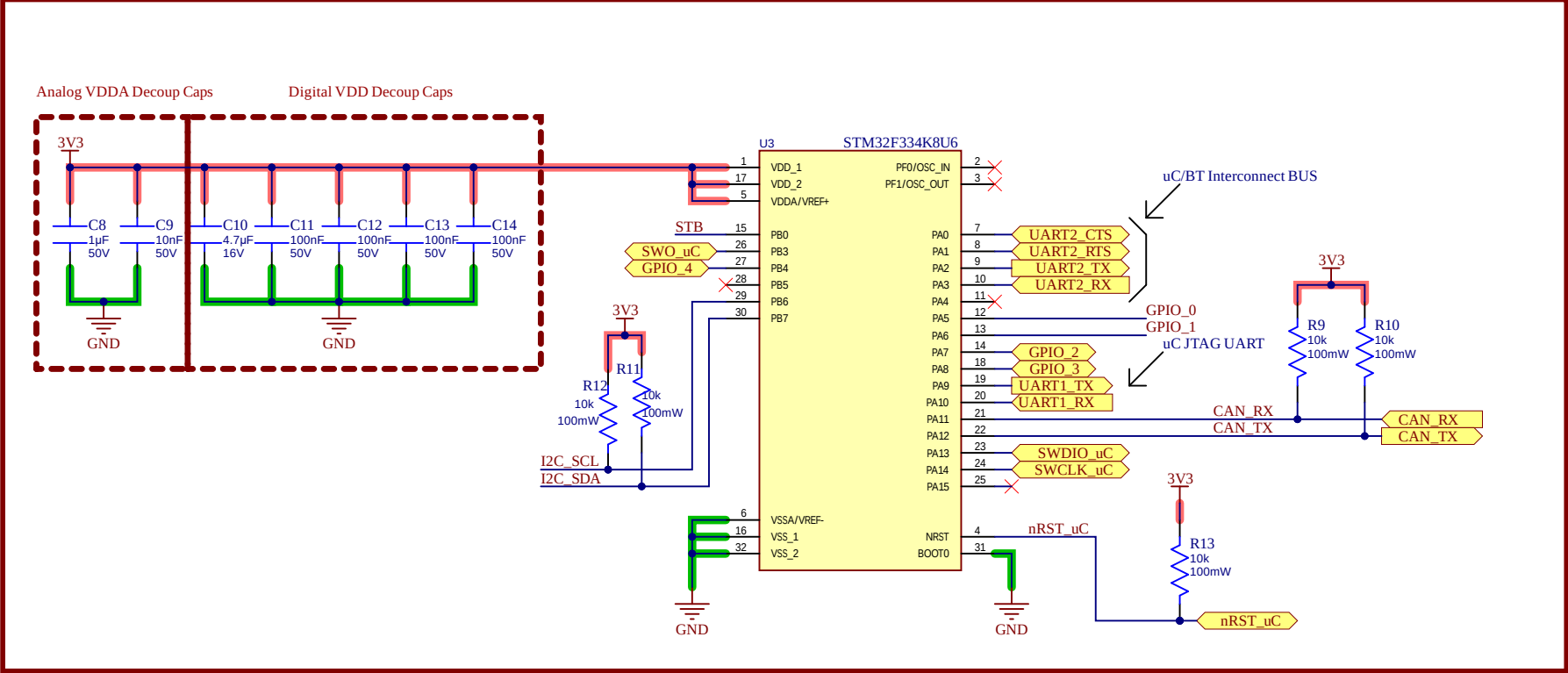
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Three circuit diagrams are shown, each representing a different LED connection:

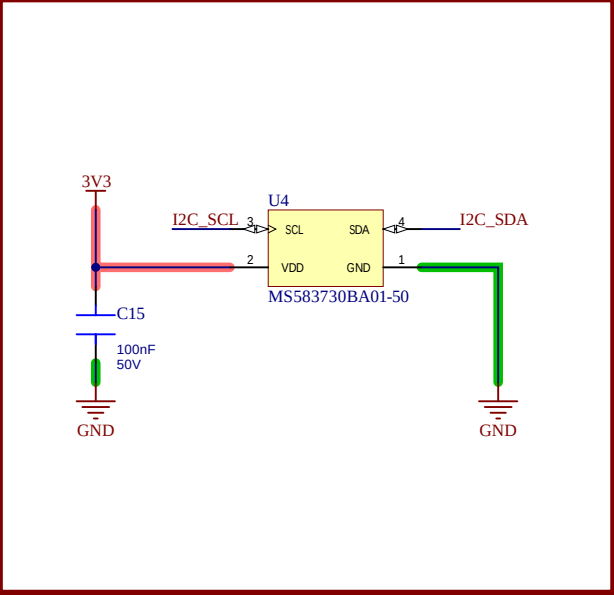
- Left Circuit:** A 3V3 supply is connected to a red resistor R2 (1k, 100mW). The other end of R2 is connected to the anode (pin 1) of LED D3 (AP1608SGC). The cathode (pin 2) of D3 is connected to GND.
- Middle Circuit:** A 5V0 supply is connected to a blue resistor R3 (2k, 100mW). The other end of R3 is connected to the anode (pin 1) of LED D4 (AP1608SGC). The cathode (pin 2) of D4 is connected to GND.
- Right Circuit:** A 12V0 supply is connected to an orange resistor R4 (4.7k, 100mW). The other end of R4 is connected to the anode (pin 1) of LED D5 (AP1608SGC). The cathode (pin 2) of D5 is connected to GND.

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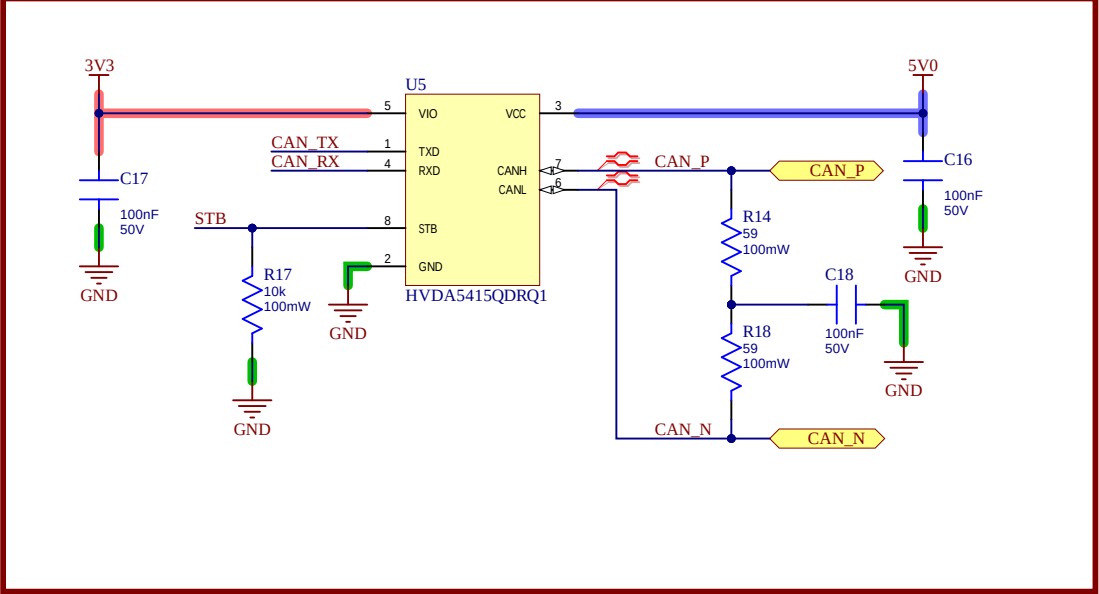
uController (STMF334)



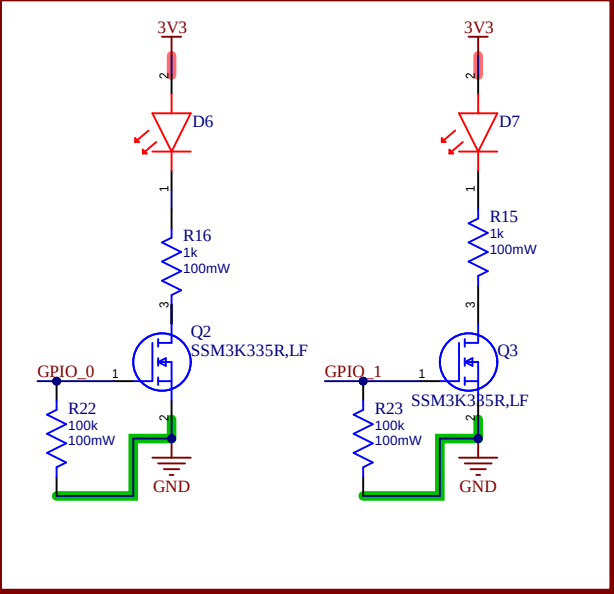
Pressure Sensor



CAN Transceiver

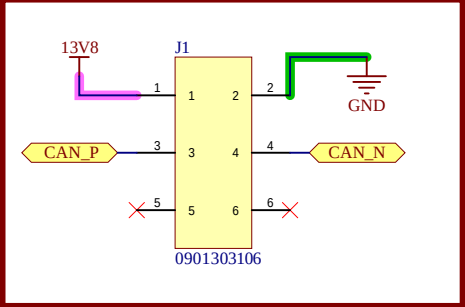


Debug/GPIO LEDS

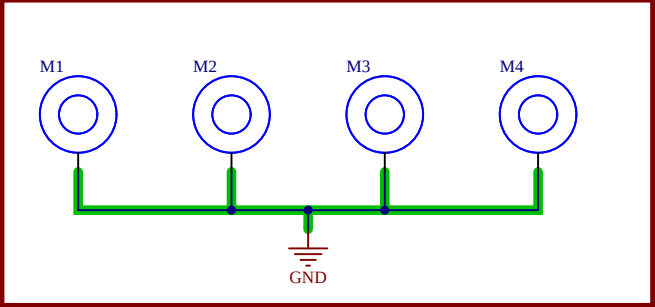


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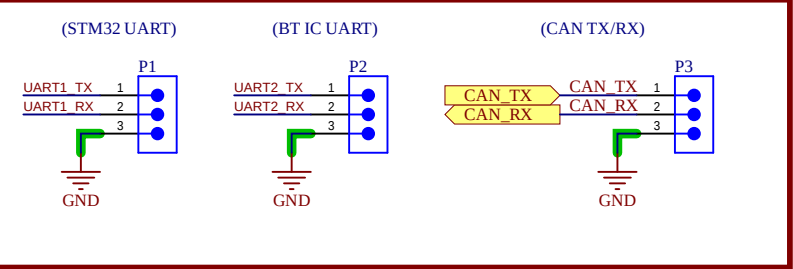
Main Connector



Mounting Holes



Debug Headers



Microcontroller JTAG Header

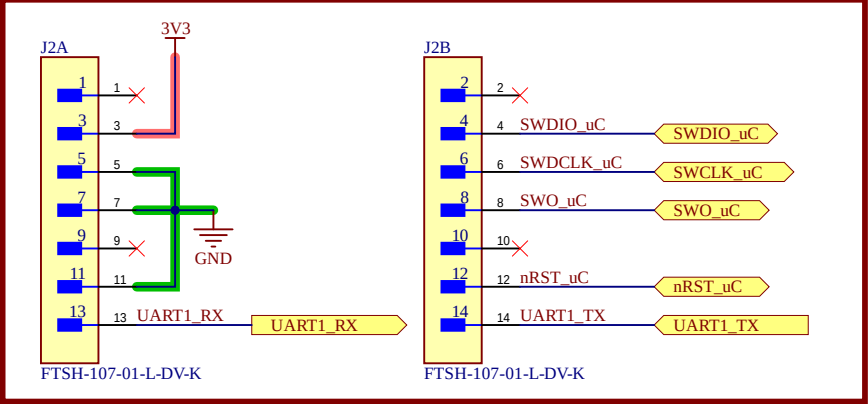
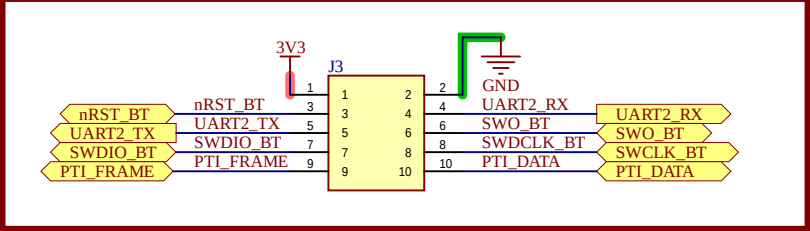


Table 6. STDC14 connector (CN1) pinout

Pin number	Description	Pin number	Description
1	Reserved ⁽¹⁾	2	Reserved ⁽¹⁾
3	T_VCC ⁽²⁾	4	T_JTMS/T_SWDIO
5	GND	6	T_JCLK/T_SWCLK
7	GND	8	T_JTDO/T_SWO ⁽³⁾
9	T_JRCLK ⁽⁴⁾ /NC ⁽⁵⁾	10	T_JTDI/NC ⁽⁵⁾
11	GNDDetect ⁽⁶⁾	12	T_NRST
13	T_VCP_RX ⁽⁷⁾	14	T_VCP_TX ⁽²⁾

1. Do not connect to the target.
2. Input for STLINK-V3SET.
3. SWO is optional, required only for Serial Wire Viewer (SWV) trace.
4. Optional loopback of T_JCLK on the target side, required if loopback is removed on the STLINK-V3SET side.
5. NC means not required for the SWD connection.
6. Tied to GND by STLINK-V3SET firmware. It might be used by the target to detect the tool.
7. Output for STLINK-V3SET

Bluetooth IC JTAG Header



Optional uC/BT nRST Tie

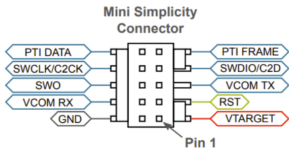
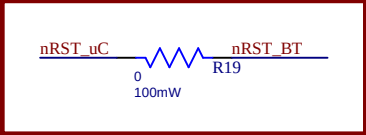


Figure 2.4. Mini Simplicity Connector

Table 2.1. Mini Simplicity Connector Pin Descriptions

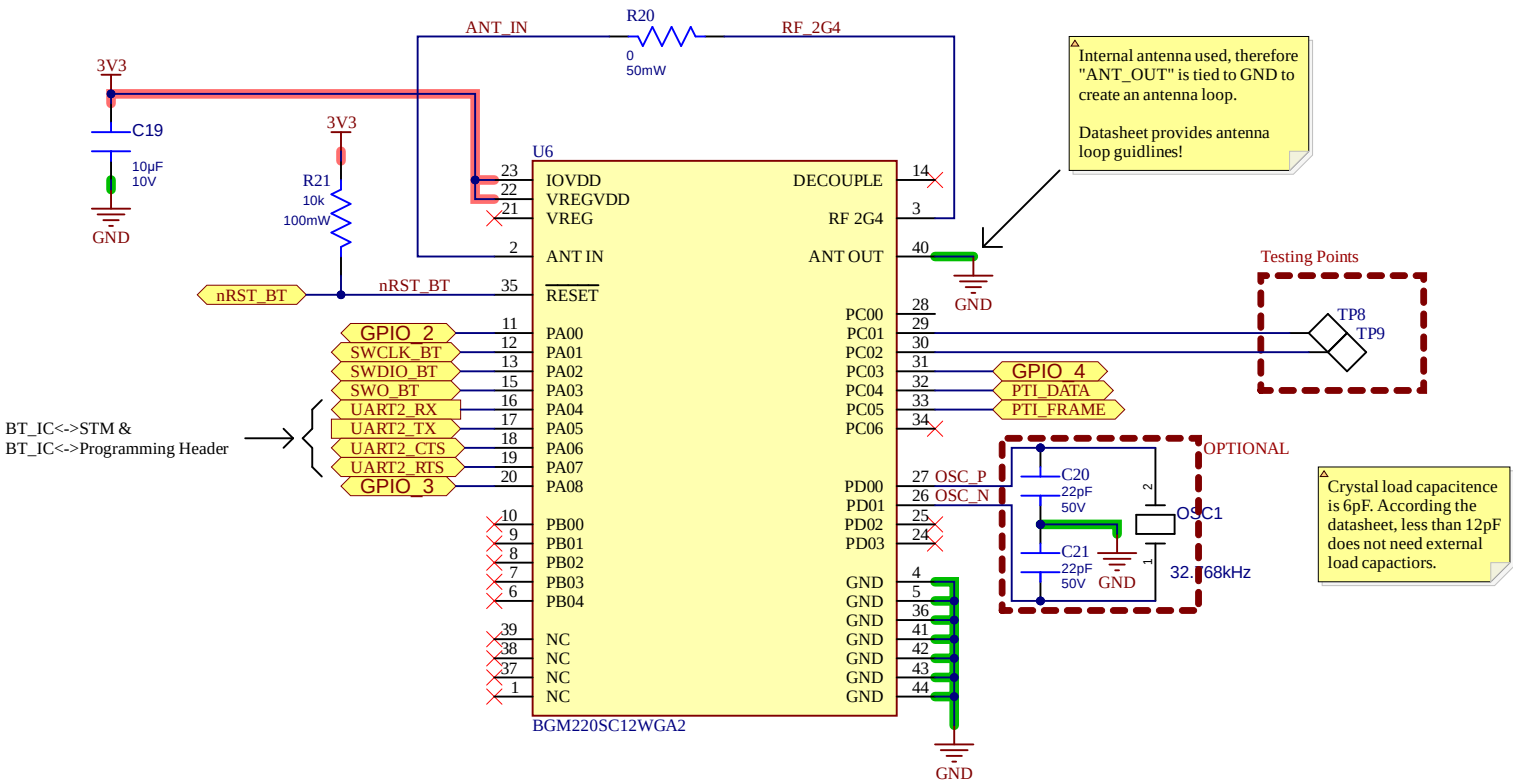
Pin Number	Function	Description
1	VTARGET	Target voltage on the debugged application. Monitored or supplied when power switch is toggled
2	GND	Ground
3	RST	Reset
4	VCOM_RX	Virtual COM Rx
5	VCOM_TX	Virtual COM Tx
6	SWO	Serial Wire Output
7	SWDIO/C2D	Serial Wire Data, alternatively C2 Data
8	SWCLK/C2CK	Serial Wire Clock, alternatively C2 Clock
9	PTI_FRAME	Packet Trace Frame Signal
10	PTI_DATA	Packet Trace Data Signal

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Wireless IC Schematic BGM220SC12WGA2

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